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Computer Vision-Based Intelligent Image Analytics System for Real-Time Product Defect Detection and Quality Monitoring

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

ITA0512 – Computer Vision

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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Under the Supervision of

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Computer Vision-Based Intelligent Image Analytics System for Real-Time Product Defect Detection and Quality Monitoring” has been carried out by G.Tejasree (Reg. No. 192511204) under the supervision of Dr. Christy Melwyn and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech ARTIFICIAL INTELLIGENCE AND DATA SCIENCE program at Saveetha Institute of Medical and Technical Sciences, Chennai.

The work presented in this report corresponds to the individual responsibility assigned to the student, with particular emphasis in Module 1: Image Acquisition & Preprocessing.

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DECLARATION

I, G.Tejasree of the ARTIFICIAL INTELLIGENCE AND DATA SCIENCE, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Computer Vision-Based Intelligent Image Analytics System for Real-Time Product Defect Detection and Quality Monitoring” is the result of my own bonafide efforts for the assigned individual contribution. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.

All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-aided tools used during the project have been appropriately disclosed.

Place: Chennai

Date: September 2026

Signature of the Student

G.Tejasree
(192511204)



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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
G.Tejasree 192511204	Module 1 – Image Acquisition & Preprocessing; preparation of the input-image pipeline; image collection, cleaning, resizing, normalization, enhancement and segmentation planning.	Designed the Module 1 workflow from image capture/ collection through preprocessing and prepared the cleaned, standardized image output required by the defect-detection stage.	Planned verification of image quality, consistency, noise removal, dimensions, normalization and enhancement; analyzed how preprocessing conditions affect downstream inspection.	Prepared Module 1 documentation and contributed to Chapters 1– 5, figures/ tables related to the preprocessing pipeline, testing discussion, limitations, future scope and references.	100%

Individual-report note: Student 2, Student 3 and Student 4 entries are intentionally removed because this report is prepared as the student's individual report. The project itself contains three modules, while this report concentrates mainly on the student's assigned Module 1.

Module ownership: Module-1 Image Acquisition & Preprocessing.

Evidence basis: The project PPT defines Module 1 as image capture/collection, noise and background removal, resizing and normalization, image enhancement through filtering/contrast adjustment/segmentation, and preparation of optimized images for AI-based defect detection.

ABSTRACT

Manufacturing quality inspection can become difficult when products must be checked continuously and consistently at production speed. Manual inspection is time-consuming, labor-intensive, and susceptible to human error, while small defects such as scratches, cracks, dents, stains, missing parts, and other surface abnormalities may be difficult to identify consistently. The proposed Computer Vision-Based Intelligent Image Analytics System ~~askes~~ this engineering problem through an automated image-based quality inspection pipeline.

The overall system captures product images, preprocesses them, extracts relevant visual information, detects defects using Computer Vision and AI/deep-learning ~~ntiques~~, and classifies products according to quality status. Within this project, the individual contribution is focused on Module 1: Image Acquisition & Preprocessing. This module establishes the quality of the input supplied to subsequent defect-detecti~~on~~ stages. Product images are collected from the production or inspection environment using cameras or imaging devices. The images are then prepared by removing noise and unwanted background information, resizing and normalizing them for consistent processing, and applying enhancement operations such as filtering, contrast adjustment, and segmentation.

The engineering significance of Module 1 is that inconsistent illumination, background content, image dimensions, noise, and contrast can make later visu~~al~~ analysis less reliable. Therefore, the module aims to produce clean and optimized product images that are suitable for AI-based defect detection. The complete project architecture further includes Module 2 for AI-based defect detection and Module 3 for prod~~uct~~ classification and quality monitoring, including dashboard-based visualization of quality trends.

The supplied project material does not report numerical experimental accuracy, latency, or a completed production deployment. Accordingly, this rep~~orts~~ not fabricate such values. Instead, it defines the verification criteria and engineering analysis needed to validate the preprocessing stage and explains its expected contribution to reliable downstream inspection. The principal outcome is a clearly d~~ef~~ined preprocessing pipeline that provides standardized image inputs for automated quality inspection.

Keywords: Computer Vision, Image Acquisition, Image Preprocessing, Defect Detection, Quality Monitoring, Image Enhancement

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	—
CV	Computer Vision	—
CNN	Convolutional Neural Network	—
ROI	Region of Interest	—
IoT	Internet of Things	—
RGB	Red-Green-Blue image representation	—
MVTec AD	MVTec Anomaly Detection dataset	—
FPS	Frames Per Second	frames/s
px	Pixel	pixel

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Computer vision allows a computer system to interpret visual information from product images and support automated inspection. In a manufacturing environment, reliable image input is essential because later analysis depends on what the camera captures. Module 1 therefore concentrates on obtaining useful product images and preparing them in a consistent form before they are passed to the defect-analysis stage.

The complete project is organized as a sequence of connected stages. Product images are first acquired or collected, then cleaned and standardized through preprocessing. The prepared images are subsequently used for visual defect analysis and classification, followed by quality-monitoring functions. This staged structure gives each module a defined responsibility and makes the overall system easier to verify and maintain.

1.2 Need for the Project

Manual inspection requires operators to repeatedly examine products and can become difficult when inspection must continue for long periods or at high production rates.

Human assessment can vary between operators and can be affected by fatigue, attention, experience, and working conditions.

Very small, low-contrast, or complex defects may be overlooked when they are judged only through direct visual inspection.

Conventional inspection procedures may not provide the speed and consistency needed for continuous monitoring of a production process.

A consistent automated image pipeline can reduce variation in the inputs and provide better-prepared images for the later AI-based inspection stage.

1.3 Problem Statement

The engineering problem is to create an image-processing pipeline that can accept product images from an inspection environment and convert them into standardized, analysis-ready inputs. The pipeline must handle common issues such as image noise, unwanted background, different image dimensions, and unsuitable contrast while retaining visual information that may indicate a defect.

1.4 Project Aim

The aim is to develop a Computer Vision and Image Analytics system for real-time product defect detection and quality monitoring, with this individual contribution concentrating on image acquisition and preprocessing so that reliable inputs are available for downstream analysis.

1.5 Project Objectives

1. Establish a suitable method for acquiring or collecting product images through cameras or other imaging devices.
2. Reduce noise and irrelevant background content while preserving important product details.
3. Convert images to consistent dimensions and normalize their representation for later processing.
4. Use appropriate enhancement operations, including filtering, contrast adjustment, and segmentation, when these improve useful visual information.
5. Produce clean and suitably prepared product images that can be passed to an AI-based defect-detection stage.
6. Define practical checks for image quality and preprocessing consistency without reporting performance measurements that have not been experimentally obtained.

1.6 Scope

The individual module covers image acquisition or collection, noise and background reduction, resizing, normalization, filtering, contrast adjustment, segmentation where useful, and verification of the resulting image quality. The output is intended to form a stable input interface for the next module.

The individual scope does not include building the complete defect-detection model, implementing the final quality dashboard, or developing physical automated rejection hardware. These functions belong to later stages of the overall project.

The design assumes that suitable product images can be obtained, that the product occupies a useful portion of the captured frame, that preprocessing operations can be applied consistently, and that the processed image can be transferred to the next project module.

1.7 Expected Engineering Outcome

The expected engineering outcome is a repeatable Module 1 workflow that transforms raw product images into cleaner, consistently sized, normalized, and appropriately enhanced images. These outputs should retain relevant product information and provide a dependable starting point for AI-based defect analysis.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was based on established work in computer vision, digital image processing, deep learning, object detection, and industrial anomaly detection. The reference material includes Szeliski, Gonzalez and Woods, Goodfellow and colleagues, YOLO, ResNet, and the MVTec AD dataset, providing a technical foundation for the proposed image-analysis workflow.

2.2 Review of Existing Work

Digital image-processing methods provide the basic operations needed to prepare inspection images. Filtering can reduce unwanted variations, contrast adjustment can make relevant structures easier to distinguish, segmentation can emphasize useful regions, and normalization can make inputs more consistent. These operations form the main technical basis of Module 1.

After preprocessing, computer-vision and deep-learning approaches can analyze the prepared images for visual patterns associated with defects. The project references YOLO for real-time object detection, ResNet for learned visual representation, and MVTec AD as an example of industrial anomaly-detection data. Module 1 supports these later methods by supplying standardized image inputs.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual visual inspection	Direct human judgment; no preprocessing pipeline required	Time-consuming; labor-intensive; human error and inconsistency	Motivates automated acquisition and preprocessing
Basic image processing	Simple and interpretable operations; useful for noise/contrast/background handling	May not identify complex defect semantics	Directly supports Module 1
Deep-learning inspection	Can learn complex visual patterns and automate defect identification	Needs suitable data and computational resources	Consumes the standardized images produced by Module 1
Proposed staged pipeline	Separates acquisition/preprocessing from detection and monitoring	Requires consistent imaging conditions and careful module interfaces	Provides the project's overall architecture

2.4 Research / Engineering Gap

A practical engineering gap exists between obtaining a raw industrial image and obtaining an image that is consistently suitable for automated defect analysis. Differences in illumination, camera setup, noise, background, image size, and contrast can affect downstream results. A controlled preprocessing stage is therefore needed to reduce avoidable variation before AI-based inspection.

2.5 Proposed Contribution

The individual contribution establishes the Module 1 path from image acquisition to a cleaned and standardized output. It defines how noise and background are handled, how dimensions and image values are standardized, and how enhancement or segmentation can be applied when appropriate. This creates a clear hand-off point for the defect-detection module.

2.6 Technical Basis for Module 1

Acquisition: obtain product images from the inspection environment using a camera or another suitable imaging source.

Noise reduction: suppress unwanted variations while protecting visual details that may represent defects.

Background handling: minimize irrelevant surroundings so the product area receives greater emphasis during analysis.

Resizing: convert images to a common target size for consistent downstream processing.

Normalization: standardize the image representation so equivalent inputs are handled consistently.

Enhancement: apply filtering and contrast adjustment selectively to make useful structures clearer.

Segmentation: isolate or emphasize the product region when separation from the background improves analysis.

Output: deliver a clean, standardized image that is ready for the AI-based defect-analysis stage.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Requirement Type	Requirement	Measurable / Verifiable Target
Functional	Acquire/collect product images	Images available in a consistent digital format
Functional	Remove noise and unwanted background	Output visibly cleaner than input and retains product region
Functional	Resize images	All processed images follow the selected target dimensions
Functional	Normalize images	Consistent input representation for downstream processing
Functional	Enhance images	Filtering/contrast adjustment improves useful visual information without destroying defect evidence
Functional	Segment relevant regions where required	Product/ROI can be distinguished for analysis
Non-Functional	Consistency	Same preprocessing sequence is applied to equivalent inputs
Non-Functional	Traceability	Input and output stages can be identified during testing
Non-Functional	Scalability	Pipeline can accept additional product images without redesigning the complete system

3.2 Constraints & Applicable Standards

The supplied project material does not identify a mandatory industrial standard or regulatory code for the prototype. Accordingly, this report avoids claiming compliance with a specific standard and instead focuses on the engineering requirements and verification conditions defined for Module 1.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Image acquisition	Clear product image with useful product region	image	High
Image dimensions	Consistent target dimensions after resizing	px	High
Noise	Reduced without removing defect information	qualitative	High
Contrast	Improved where low contrast hides relevant	qualitative	Medium

Background	features	qualitative	High
Normalization	Consistent image representation	qualitative	High
Segmentation	Relevant product region isolated/emphasized when needed	qualitative	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Manual image preparation. An operator selects and adjusts images before inspection. Although straightforward, this approach depends heavily on human effort and is difficult to maintain consistently as the number of images increases.

Alternative 2: Fixed preprocessing pipeline. Each image passes through a defined sequence covering acquisition, noise/background handling, resizing, normalization, and enhancement. This approach provides repeatability and makes individual processing stages easier to inspect and test.

Alternative 3: End-to-end AI processing from raw images. The model receives images with limited preprocessing and learns more of the required representation itself. This can reduce explicit preprocessing, but it increases dependence on suitable training data and model development.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Consistency	5	2	5	4
Processing control	4	3	5	3
Scalability	4	2	5	4
Ease of verification	4	3	5	3
Dependence on training data	3	5	4	2

3.5 Selected Design & Detailed Design

Alternative 2 is selected because it matches the responsibilities assigned to Module 1 and creates a clear, testable boundary before Module 2. Separating preprocessing from detection also makes it possible to inspect image quality before inference and to adjust individual operations without redesigning the complete system.

Figure 3.1 – Module 1 engineering workflow

Stage	Input	Operation	Output
1	Camera / collected image	Capture or collect product image	Raw product image
2	Raw product image	Noise and background handling	Cleaner product image
3	Cleaner image	Resize and normalize	Standardized image
4	Standardized image	Filtering / contrast adjustment / segmentation	Enhanced image
5	Enhanced image	Quality verification	Optimized image ready for Module 2

Systems approach: Module 1 accepts images from the production or inspection environment and converts them into a standardized representation. Module 2 then uses these prepared images for AI-based defect analysis, while Module 3 uses downstream results for quality monitoring. The interface between stages is therefore an important part of the design.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Explicit preprocessing pipeline	End-to-end raw-image model	Clear control and verification of input quality	Adds preprocessing steps	Selected because Module 1 is explicitly defined in the project architecture
Image resizing	Allow arbitrary image sizes	Consistent downstream input	May alter very fine details if poorly chosen	Selected as required by PPT; target dimensions must preserve relevant features
Contrast enhancement	No enhancement	Can improve visibility of subtle patterns	Over-enhancement can distort evidence	Selected conditionally based on image quality
Segmentation	No segmentation	Preserves full image	Background may distract analysis	Selected when background/product separation is beneficial

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Avoid unnecessary repeated processing and manual rework	Modular preprocessing and automated inspection objective	Use a repeatable pipeline so images are prepared consistently
Ethics	Avoid manipulating images in a way that hides genuine defects	Need to preserve defect evidence	Enhancement must improve clarity without deleting relevant features
Health & Safety	Reduce dependence on continuous manual visual inspection	Project objective includes reducing human inspection effort	Automation supports safer, more consistent inspection workflows
Security	Protect inspection images and results from unauthorized access	Images may represent production information	Use controlled storage/access when deployed
Risk	Lighting/camera variation	Listed project limitation	Verification must include varied acquisition conditions

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Poor lighting	Medium	High	High	Control lighting and verify image quality	Medium
Camera variation	Medium	Medium	Medium	Standardize camera setup and preprocessing	Low–Medium

Noise removal removes fine defects	Medium	High	High	Validate before/ after images against defect visibility	Medium
New product type	High	Medium	High	Reconfigure/retrain downstream system as required	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The implementation follows the modular structure described in the project material. Module 1 begins with image acquisition or collection and then applies the required preparation stages: cleaning, background handling, resizing, normalization, and suitable enhancement. The resulting image is intended to be passed to the AI detection stage without introducing unsupported assumptions about measured system performance.

Resource / Tool	Purpose	Stage Used
Camera / imaging device	Capture product images	Acquisition
Image dataset / collected product images	Provide inspection inputs	Acquisition and testing
Computer Vision / image-processing methods	Clean, resize, normalize and enhance images	Preprocessing
AI / Deep Learning methods	Downstream defect detection	Module 2 interface
Dashboard / visualization	Downstream quality monitoring	Module 3 interface

4.2 Software / Implementation

Module 1 is organized as a sequential image-processing workflow. A product image is first obtained from a camera or inspection source. Unwanted noise and background content are then reduced, after which the image is resized and normalized. Filtering, contrast adjustment, and segmentation are applied when they are useful for preserving or emphasizing relevant visual information. The final output is checked for suitability before it is handed to the defect-analysis stage.

Figure 4.1 – Module 1 implementation workflow

Input	Processing	Verification Point	Output
Raw image	Noise/background reduction	Product information retained	Cleaner image
Cleaner image	Resize	Target dimensions consistent	Standardized image
Standardized image	Normalize	Consistent representation	Normalized image
Normalized image	Enhance / segment	Useful features remain visible	Optimized image

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion	Specification Required Value	Achieved Value / Status
Image acquisition	Inspect sample image set	Product is visible and usable for processing	Usable product images	To be verified on implementation; no numeric result in PPT
Noise reduction	Compare input and output images	Noise is reduced without removing relevant product detail	Cleaner image	Qualitative verification required
Background reduction	Visual/ROI inspection	Irrelevant background is reduced	Product region remains usable	Qualitative verification required
Resizing	Check image dimensions	All outputs match selected target dimensions	Consistent dimensions	Implementation check required
Normalization	Check processed representation	Equivalent inputs are consistently represented	Consistent normalized input	Implementation check required
Enhancement	Before/after visual inspection	Useful contrast/detail is improved without distortion	Improved visual quality	Qualitative verification required
Segmentation	Inspect ROI/product separation	Relevant region is separated/emphasized where needed	Useful segmentation	Condition-dependent
Module interface	Pass output to detection stage	Output is accepted by next stage	Analysis-ready image	Interface verification required

4.4 Results

The supplied project material defines the intended Module 1 functions but does not provide measured accuracy, latency, confusion matrices, or completed production-deployment results. Therefore, the result reported here is an engineering verification framework rather than a fabricated numerical performance claim. Actual measurements should be added after the implementation is executed and tested with real image samples.

4.5 Data Analysis & Engineering Judgment

Preprocessing should be judged by whether important product and defect information remains usable, not simply by whether the processed image looks visually attractive. Excessive filtering can remove small defect patterns, while strong contrast enhancement can introduce artificial edges. Background reduction and segmentation should improve focus on the product without cutting away relevant regions. Resizing and normalization should also maintain sufficient spatial information for downstream detection.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design an automated computer-vision system for defect detection	Overall three-module architecture in PPT	Partially evidenced in supplied material
Acquire/collect product images	Module 1 explicitly includes capture and collection	Defined / implementation evidence required
Remove noise/background	Module 1 explicitly includes both	Defined / implementation evidence required
Resize and normalize images	Module 1 explicitly includes both	Defined / implementation evidence required
Apply filtering, contrast adjustment and segmentation	Module 1 explicitly includes enhancement methods	Defined / implementation evidence required
Prepare images for AI defect detection	Module 1 output is clean and optimized images ready for analysis	Defined
Provide real-time defect detection and monitoring	Modules 2 and 3 define these functions	Outside individual Module 1 scope; system-level implementation evidence required

4.7 Strengths & Limitations

Strengths:

Clear separation between image preparation, defect detection, and quality-monitoring stages makes the system easier to organize and test.

Module 1 establishes a defined transition from raw image acquisition to a standardized input for AI-based inspection.

The preprocessing sequence explicitly covers noise and background reduction, resizing, normalization, filtering, contrast adjustment, and segmentation where appropriate.

Image quality can be checked before the prepared image is supplied to the downstream inference stage.

Limitations:

The available project material contains no numerical validation results for the implemented pipeline.

Changes in lighting and camera configuration can alter the quality and consistency of captured images.

Very small or visually complex defects may be difficult to preserve during preprocessing or recognize in later stages.

Real-time operation may require additional computational resources depending on the image size and processing method.

New product categories may require parameter changes or retraining in the downstream inspection stages.

The complete inspection system can still produce false-positive or false-negative decisions and

therefore requires validation.

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
G.Tejasree (192511204)	Module 1 – Image Acquisition & Preprocessing	Individual ownership of Module 1 design, preprocessing workflow, verification planning and report documentation
Item	Estimated Cost	Actual Cost
Camera / imaging source	Not specified in PPT	Not specified in PPT
Computing resources	Existing academic resource	Not specified in PPT
Dataset / image collection	Not specified in PPT	Not specified in PPT
Software / processing tools	Not specified in PPT	Not specified in PPT

Budget note: The supplied project material does not specify project costs. No cost values have been invented in this report.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project presents a Computer Vision-Based Intelligent Image Analytics System intended for real-time product defect detection and quality monitoring. The architecture separates image preparation, AI-based defect analysis, and quality-monitoring functions so that each stage has a clear engineering responsibility.

Module 1 provides the foundation by acquiring or collecting product images and preparing them through noise and background reduction, resizing, normalization, filtering, contrast adjustment, and segmentation when appropriate. Its purpose is to deliver consistent, analysis-ready images to the next stage.

Because the supplied material does not include measured accuracy, processing time, or production-deployment evidence, no unsupported numerical performance is claimed. The main outcome of this module is a defined and verifiable preprocessing workflow that can be evaluated using actual implementation data.

5.2 Future Work

- Deploy the image-acquisition workflow with suitable cameras on representative production lines.
- Test preprocessing performance under different lighting levels, camera positions, and imaging conditions.

- Tune enhancement and segmentation settings for different product categories and defect characteristics.

- Investigate edge-AI deployment to reduce processing delay and support faster inspection.

- Connect inspection outputs with IoT-enabled smart-factory monitoring systems.

- Extend the system with automated rejection mechanisms using suitable robotic or conveyor equipment.

- Develop predictive quality analysis to identify process conditions that may contribute to future defects.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Learn how image acquisition forms the first dependable stage of an automated computer-vision inspection workflow.

- Build practical understanding of noise reduction, background handling, resizing, normalization, and image enhancement.

- Understand that preprocessing decisions can directly affect the quality of later defect-detection results.

Recognize practical industrial issues such as lighting variation, camera differences, processing requirements, and inspection consistency.

Engineering & Professional Skills Developed:

Develop modular thinking by defining responsibilities and interfaces between connected system stages.

Develop the ability to prepare verification plans from explicit engineering requirements.

Improve technical documentation and use evidence-based reasoning when describing system behavior.

Learn to identify limitations and risks clearly instead of presenting unsupported performance claims.

Individual Reflection – G.Tejasree: The main engineering challenge in Module 1 is to remove unwanted variation without losing the visual evidence of a real defect. Working through the acquisition and preprocessing stages helped me understand that image quality is not only about making a picture look clearer; every processing choice can influence what the next AI stage is able to recognize. I also learned the importance of consistent dimensions, normalization, controlled enhancement, verification, and honest reporting of limitations. This module strengthened my understanding of how a well-defined preprocessing stage supports the reliability of the complete computer-vision inspection system.

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APPENDICES

Appendix A – Detailed Module 1 Workflow

Step	Description	Expected Output
A1	Capture/collect product image	Raw image
A2	Inspect image quality and remove obvious unwanted background/ noise	Cleaner image
A3	Resize image to consistent dimensions	Standard-size image
A4	Normalize image representation	Normalized image
A5	Apply filtering and contrast adjustment where needed	Enhanced image
A6	Apply segmentation where appropriate	Relevant product/ROI
A7	Verify image quality before hand-off	Optimized image ready for Module 2

Appendix B – Engineering Drawings / Schematics

Recommended report figure: a horizontal pipeline showing Camera / Image Source → Image Acquisition → Noise & Background Removal → Resize → Normalize → Filtering / Contrast Adjustment → Segmentation → Verification → Module 2 AI Defect Detection.

Appendix C – Source Code / Code Repository Information

Complete source code was not included in the supplied project material. If an approved repository or implementation package is available, its verified location should be inserted here. Only actual code and approved repository information should be reported.

Appendix D – Experimental / Raw Data

The supplied project material does not contain raw experimental data. Actual before/after image samples, processing-time measurements, and downstream model results should be added only after the corresponding implementation tests have been completed.

Appendix E – Individual Contribution Evidence

Student: G.Tejasree (192511204)

Assigned module: Module 1 – Image Acquisition & Preprocessing.

Image acquisition and collection workflow.

Noise and unwanted background reduction.

Image resizing and normalization.

Filtering and contrast adjustment.

Segmentation planning where appropriate.

Preparation of clean and optimized images for AI-based defect detection.

Testing and analysis plan for preprocessing quality.

Technical report preparation for the individual contribution.

Appendix F – Capstone Project Outcome Mapping

Outcome / Area	Evidence Location
Complex engineering problem formulation	Chapter 1, Chapter 2
Engineering design under constraints	Chapter 3
Written/oral technical communication	Whole report
Ethics and professional responsibility	Chapter 3.7
Teamwork and leadership	Chapter 4.8
Experimentation, analysis and engineering judgment	Chapter 4.3–4.6
Independent acquisition of new knowledge	Chapter 5.3
Sustainability and societal/environmental impact	Chapter 3.7
Risk assessment and mitigation	Chapter 3.7
Security considerations	Chapter 3.7
Integrated/system-level engineering	Chapter 3.5
Project planning and management	Chapter 4.8
Individual contribution	Contribution Statement + Chapter 4.8